Muhammad Fasih Ullah Farooqui
BSCS 6-C
23122119
ARTIFICIAL INTELLIGENCE
MISS UZMA

AI Agent Formulation & Evaluation Assignment

Course Title: Artificial Intelligence
Agent Name: SmartThermo AI (Automated Climate Control System)
Date: September 5, 2026

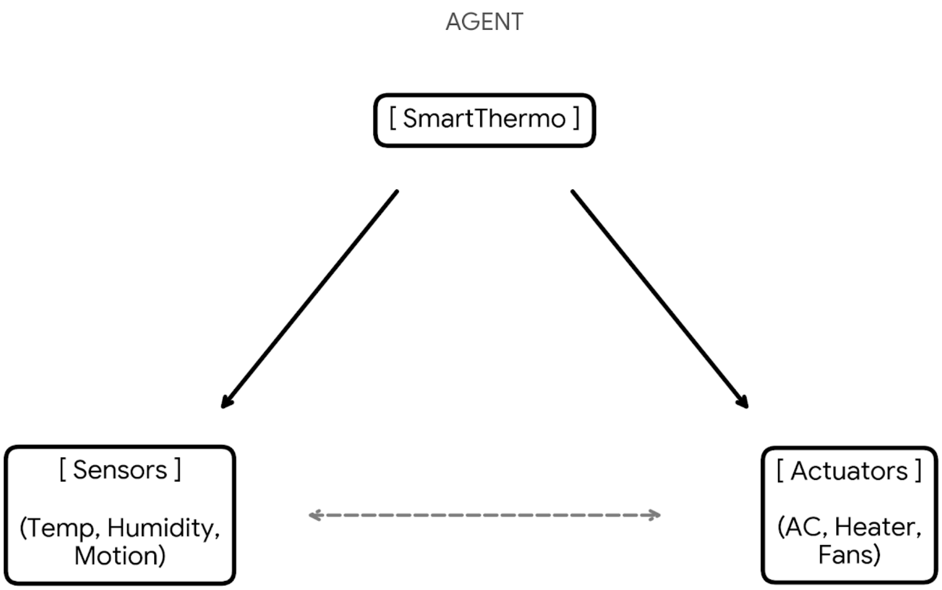
1. AI Agent Selection & Name

Agent Name: SmartThermo AI

Introduction
An AI agent basically takes inputs from its environment through sensors and performs actions using actuators to solve a specific problem. In smart homes, an automated thermostat is a classic practical example of a software agent. SmartThermo AI tracks room temperature, humidity, and movement in the house to automatically turn the heating or cooling on and off, keeping the room comfortable without needing constant manual adjustments.

2. Agent Formulation

• Graphical Representation



• Textation

1. **Sensor:** Collects room temperature, indoor humidity levels, and room motion occupancy from environmental sensors.
2. **Actuator:** Executes physical adjustments by turning on/off the AC, heater compressor, or fan speed based on comfort rules.
3. **Environment:** Monitors the indoor home climate stream to maintain optimum target temperature while conserving energy.

3. PEAS Framework Evaluation

PEAS	Description
P	Temperature Accuracy & Energy Efficiency Optimization
E	Indoor Home Rooms & Climate Control System
A	AC Compressor, Heater Switch & Fan Controller
S	Temperature Sensor, Humidity Gauge & Motion Sensor

4. Environment Type

Property	Type
Observability	Partially
Determinism	Stochastic
Episodes	Sequential
Change	Dynamic
Values	Continuous
Agents	Single

Conclusion

SmartThermo AI is a great example of how simple AI agents handle changing real-world conditions. Breaking down its components through the PEAS framework helps us understand how sensors and actuators interact to automate everyday home tasks effectively.